

Module 3: Assignment

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IFT 372: Wireless Fundamentals

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Questions

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a) How can the LTE core network be interconnected with legacy core networks?

LTE core networks have a few features that allow for compatibility with legacy core networks. LTE supports dual-stack, which allows them to be connected with GSM and UMTS since those technologies only support IPv4. Interfaces like S3 and S4 allow for seamless handovers between LTE and legacy systems. Fallbacks such as CSFB and SRVCC allow for voice calls to fall back to legacy systems without disruption to the user.

b) Why should this be done?

This should be done to ensure backward compatibility. Not all wireless networks are based on LTE, some are based on legacy systems like UMTS or GSM. Traffic should not be confined to one technology or another. This would defeat the purpose of an internetwork, which has the goal of global interconnectivity. Wireless networks need an interface through which they can smoothly pass data for seamless global communication.

2) Describe different options for the backhaul connection of the eNB.

eNB's have a few options for backhaul connections. They can have wired backhaul's including fiber and copper. Fiber offers high bandwidth connections with low latency but tends to be costly. Copper supports less bandwidth and higher latency than fiber but is much cheaper. There are also wireless options such as millimeter wave and satellite. Millimeter-wave connections offer high bandwidth but come at the cost of relatively short range and can be easily obstructed

by objects. Satellite connections are good for locations with unforgiving terrain but come at the cost of slower speeds with higher latency.

3) Describe the differences between the tasks for the MME and the tasks of the S-GW.

The MME and S-GW work in tandem to facilitate wireless communication. The MME handles tasks such as managing authentication, sessions, mobility, and fallback to legacy networks. The S-GW handles the routing of packets and serves as a mobility anchor for handovers.

4) How does a mobile device get access to the PUSCH?

In short, the mobile device sends a request to the eNB saying “I want access to the PUSCH”, the eNB sees what uplink resources are available and allocates them for the mobile device, the eNB then replies to the mobile with what resources it allocated for it via the PDCCH. The device is now able to transmit data on the PUSCH.

5) How is mobility controlled in RRC idle state?

In RRC idle state, mobility is mainly controlled cell selection and reselection. The device is constantly monitoring cell signal strengths from the nearest base stations. It stays connected to the strongest cell until it detects a stronger cell signal. It then selects the stronger cell signal for connection.

6) How many subcarriers are used for a 10MHz FDD-LTE channel?

600 subcarriers are used for the 10MHz FDD-LTE channel if the subcarrier spacing in the network is 15kHz. Subcarriers carry their own data stream and have specific functions. Most carry control information or user data but the rest serve other roles such as guard bands to minimize interference.

7) What are the differences between ARQ and HARQ?

ARQ implements basic error control mechanism. If a device successfully receives a packet, it will reply with an ACK. If the packet is corrupted in any way, it will send back a NAK and the sender will retransmit the packet. HARQ is a more efficient error control mechanism. It works in a similar way but requests the sender to only retransmit the parts of the packet that were corrupted on the first transmission.

8) What is an RB?

A resource block is the smallest unit of resource allocated by the eNodeB scheduler. It is specified as 12 adjacent subcarriers. The eNodeB scheduler allocates resource blocks to users depending on their data needs. Utilizing RB's helps facilitate efficient and flexible utilization of resources.

9) What is CS-fallback?

CS-fallback is the mechanism that enables LTE voice calls to swap to legacy circuit-switched systems if needed. LTE is entirely packet-switched so CS-fallback comes into play when a call needs to be handled by a circuit-switched network.

10) What is the big disadvantage of Internet-based voice services compared to network operator-based voice services?

Compared to operator-based voice services, internet-based voice services have the disadvantage of varying and overall worse audio quality. Internet-based voice relies on your internet connection, sharing bandwidth with other services for example, which causes audio quality to vary depending on external factors. A stable internet connection is needed for internet-based voice services to work reliably.

11) What is the difference between a cell change order and a handover?

The main difference between a cell change order and a handover is that a cell change order is limited to just the instruction sent from the network to the UE to connect to another cell while a handover also has to transfer an existing session seamlessly from the old cell to new cell without disruption. The handover includes the cell change order but employs other mechanisms on top of it to ensure session continuity.

12) What is the difference between a default and a dedicated bearer?

Default bearers are the initial bearer created when a user session starts. They support best-effort traffic and can require other bearers. Dedicated bearers are secondary to default bearers and facilitate different levels of QoS depending on the traffic (*Default Bearer*, n.d.).

13) What is the difference between an S1 and an X2 handover?

S1 handovers are slower than X2 handovers because S1 handovers occur between differing radio access technologies, which calls for the involvement of the EPC. X2 handovers occur in an intra-LTE manner directly between eNodeB's without the need for an EPC.

14) What is the purpose of DRX in RRC connected state?

The purpose of DRX in RRC connected state is to extend mobile device battery life. It reduces the amount of time the device needs to actively listen for downlink traffic. The mobile device negotiates specific periods of time where data can be transferred with the network and enters a low power state between those periods.

References

Default bearer. Default Bearer - an overview | ScienceDirect Topics. (n.d.).

<https://www.sciencedirect.com/topics/computer-science/default-bearer#:~:text=Default%20bearers%20support%20only%20best,bearers%20for%20specific%20traffic%20flows.&text=Dedicated%20Bearer%3Awhich%20guarantees%20different,tunnel%20between%20UEs%20and%20PGWs.>